

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics

HSC ASSESSMENT TASK 3 JUNE 2007

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions

NAME : _____

QUESTION 1	QUESTION 2	QUESTION 3	QUESTION 4	QUESTION 5	TOTAL

Question 1 (11 marks) **Marks**

- | | |
|---|---|
| a) Evaluate e^7 giving your answer correct to 3 significant figures. | 2 |
| b) Find the exact value of $\cos \frac{5\pi}{6}$. | 1 |
| c) Find a primitive of $3e^x + \cos x$ | 2 |
| d) Convert 0.56 radians to degrees giving your answer to the nearest degree. | 1 |
| e) Find the equation of the tangent to $y = \cos \frac{x}{2}$ at the point $(\pi, 0)$. | 3 |
| f) Sketch the curve $y = 4 \cos 2x$ for $0 \leq x \leq 2\pi$. | 2 |

Question 2 (11 marks) (Start a new page) **Marks**

- | | |
|---|---|
| a) Differentiate with respect to x : | |
| i) $\tan x$ | 1 |
| ii) $\sin(x^2 + 1)$ | 2 |
| iii) $\frac{e^{2x}}{x}$ | 2 |
| b) The area bounded by $y = e^{2x}$ and the x axis from $x = 1$ to $x = 3$ | 3 |
| is rotated about the x axis. Find the volume of the solid of revolution formed. | |
| c) Find the value of k for which $y = e^{-2x}$ satisfies the equation | 3 |

$$\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + k y = 0$$

Question 3 (11 marks) (Start a new page)

Marks

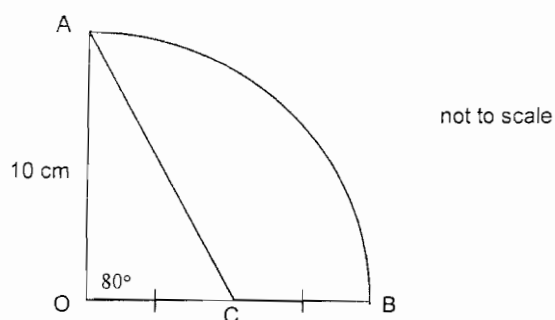
a) Solve $\tan x - 1 = 0$ for $0 \leq x \leq 2\pi$. (answer must be in radians)

2

b) Sketch the curve $y = e^x - 1$

1

c)



OAB is a sector with radius 10 centimetres.

$\angle AOB = 80^\circ$. C is the midpoint of OB .

i) Find the exact length of the arc AB .

2

ii) Find the area enclosed by the arc AB and the lines AC and CB .

3

d) i) Two values of the function $y = \frac{x^2}{x+9}$ are shown in the table.

x	0	3	6	9
y	0	0.75		

Copy the table onto your answer sheet and fill in the missing values.

1

ii) Use the trapezoidal rule with 3 intervals (4 function values) to estimate

2

$$\int_0^9 \frac{x^2}{x+9} dx$$

Question 4 (11 marks) (Start a new page)	Marks
a) i) State the period of the function $y = \sin(2x - \pi)$.	1
ii) State the amplitude of the function $y = \sin(2x - \pi)$	1
b) i) Use the standard integral table to find $\int \sec 2x \tan 2x \, dx$	1
ii) Find $\int e^{4x} + 1 \, dx$	1
c) i) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$	2
ii) Evaluate $\int_0^2 \sin \frac{\pi x}{4} \, dx$	2
d) Find the area enclosed by the curve $y = \sin x$ for $0 \leq x \leq 2\pi$, the line $y = 1$ and the y axis.	3

Question 5 (11 marks) (Start a new page)**Marks**

- a) i) On the same set of axes draw neat sketches of the functions

2

$$y = \sin x \text{ for } -2\pi \leq x \leq 2\pi \quad \text{and} \quad y = 1 - \frac{x}{4}$$

- ii) How many solutions does the equation $\sin x = 1 - \frac{x}{4}$ have ?

1

- b) The curve $y = x + \cos x$ has one stationary point for x between 0 and 2π .

4

Find this stationary point and determine its nature.

- c) i) Find $\frac{d}{dx}(xe^{2x})$

2

- ii) Use the above result to find $\int xe^{2x} dx$

2

End of Paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS - ASS TASK 3 2007

QUESTION 1

a. 1100

b. $-\frac{\sqrt{3}}{2}$

c. $3e^x + \sin x$

d. 32°

e. $y' = -\frac{1}{2} \sin \frac{x}{2}$

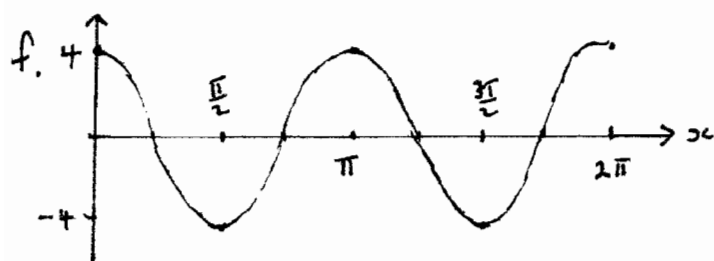
when $x = \pi$

$y' = -\frac{1}{2}$

$\therefore m_T = -\frac{1}{2} \quad (\pi, 0)$

$y - 0 = -\frac{1}{2}(x - \pi)$

$x + 2y = \pi$



QUESTION 2

a. i) $\sec^2 x$

ii) $2x \cos(x^2 + 1)$

iii) $\frac{e^{2x}(2x-1)}{x^2}$

b. $V = \pi \int_1^3 e^{4x} dx$

$= \frac{\pi}{4} [e^{4x}]_1^3$

$= \frac{\pi}{4} [e^{12} - e^4]$

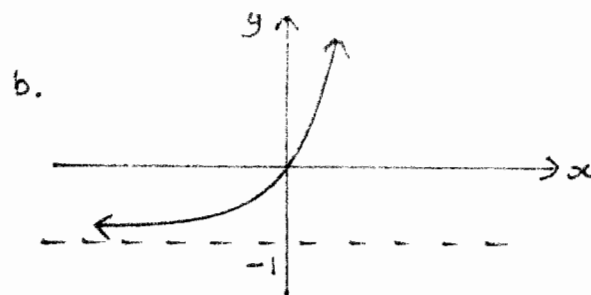
c. $y' = -2e^{-2x} \quad y'' = 4e^{-2x}$

$4e^{-2x} + 3(-2e^{-2x}) + ke^{-2x} = 0$

$e^{-2x} (4 - 6 + k) = 0$

QUESTION 3

a. $x = \frac{\pi}{4}, \frac{5\pi}{4}$



c. i) $80^\circ = \frac{4\pi}{9}$

$L = 10 \times \frac{4\pi}{9}$
 $= \frac{40\pi}{9} \text{ cm}$

ii)

$A = \frac{1}{2} \cdot 10^2 \cdot \frac{4\pi}{9} - \frac{1}{2} \cdot 10 \cdot 5 \cdot \sin 80^\circ$

$= 45.2 \text{ cm}^2$

d. i)

x	0	3	6	9
y	0	0.75	2.4	4.5

ii) $\int_0^9 \frac{x^2}{x+9} dx$

$\approx \frac{3}{2} [0 + 4.5 + 2 \times (0.75 + 2.4)]$

$= 16.2$

QUESTION 4

a. i) Period = π

ii) Amplitude = 1

b. i) $\frac{1}{2} \sec 2x + c$

ii) $\frac{1}{4} e^{4x} + x + c$

c. i) $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$

$$= \left[\frac{1}{2} \tan 2x \right]_0^{\frac{\pi}{8}}$$

$$= \frac{1}{2} \tan \frac{\pi}{4} - \frac{1}{2} \tan 0$$

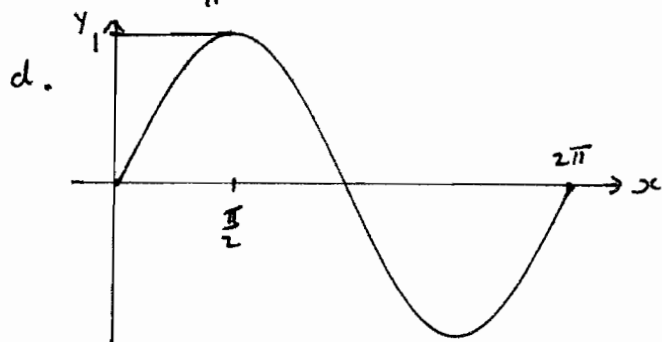
$$= \frac{1}{2}$$

ii) $\int_0^2 \sin \frac{\pi x}{4} \, dx$

$$= \left[-\frac{4}{\pi} \cos \frac{\pi x}{4} \right]_0^2$$

$$= -\frac{4}{\pi} \left[\cos \frac{\pi}{2} - \cos 0 \right]$$

$$= \frac{4}{\pi}$$



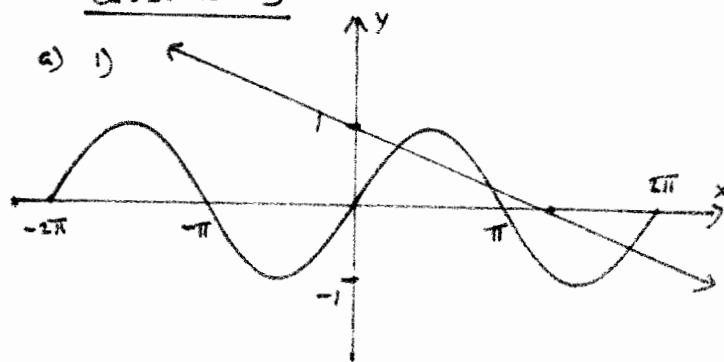
$$A = \frac{\pi}{2} - \int_0^{\frac{\pi}{2}} \sin x \, dx$$

$$= \frac{\pi}{2} - \left[-\cos x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} - \left[-\cos \frac{\pi}{2} + \cos 0 \right]$$

$$= \frac{\pi}{2} - 1 \quad \text{sq units}$$

QUESTION 5



ii) 3 solutions

b) $y = x + \cos x$

$$y' = 1 - \sin x$$

st. pts when $y' = 0$

$$\sin x = 1$$

$$x = \frac{\pi}{2}$$

$$y = \frac{\pi}{2}$$

test

x	$\frac{\pi}{2} - \text{ve}$	$\frac{\pi}{2}$	$\frac{\pi}{2} + \text{ve}$
y'	tve	0	tve

/ — /

$\therefore$ horizontal point of inflexion at $\left(\frac{\pi}{2}, \frac{\pi}{2} \right)$

c) i) $\frac{d}{dx} (x e^{2x}) = e^{2x} + 2x e^{2x}$

ii) $\int x e^{2x} \, dx$

$$= \frac{1}{2} (x e^{2x} - \int e^{2x} \, dx)$$

$$= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + c$$